Lab Nov 17

Matrices in R

Let's use our WHI experiment data

```
whi<- read.csv("whidata.csv")
whi$age50s<- ifelse(whi$age=="50-59", 1, 0)
whi$age60s<- ifelse(whi$age=="60-69", 1, 0)
whi$age70s<- ifelse(whi$age=="70-79",1,0)
```

Start with simple additive model

```
whimod1<- lm(breastcancer ~ treat + age60s + age70s, data=whi)

summary(whimod1)
```

```
Call:
lm(formula = breastcancer ~ treat + age60s + age70s, data = whi)

Residuals:
     Min       1Q   Median       3Q      Max
-0.02823 -0.02539 -0.02028 -0.02006  0.98505

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.014945   0.002281   6.552 5.85e-11 ***
treat       0.005112   0.002263   2.259  0.02391 *
age60s      0.005337   0.002585   2.065  0.03895 *
age70s      0.008168   0.003129   2.611  0.00904 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.1458 on 16604 degrees of freedom
Multiple R-squared:  0.0007668, Adjusted R-squared:  0.0005863
F-statistic: 4.247 on 3 and 16604 DF,  p-value: 0.005241
```

How can we get the same thing using matrices? We want $(X'X)^{-1}X'Y$, so we need a matrix X that has our covariates in it and a column of 1s so we can estimate an intercept.

```
df.mat<- data.frame(constant=rep(1,length(whi$breastcancer)), treat=whi$treat, age60s=whi$

X<- data.matrix(df.mat)

dim(X)
```

```
[1] 16608     4
```

To test our matrix functions, I want to use something smaller to be able to check it. I will use the matrix from PS 3: $C = \begin{bmatrix} 3 \\ 2 \\ 5 \\ 7 \end{bmatrix}$ and $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$

```
A<- matrix(c(1,1,1,1,0,1,1,2),ncol=2)
dim(A)
```

```
[1] 4 2
```

```
C<- matrix(c(3,2,5,7),ncol=1)
dim(C)
```

```
[1] 4 1
```

```
#What is the transpose of A?
t(A)
```

```
     [,1] [,2] [,3] [,4]
[1,]    1    1    1    1
[2,]    0    1    1    2
```

```r
t(A)%*%A
```

```
     [,1] [,2]
[1,]    4    4
[2,]    4    6
```

```r
#What is the inverse of A'A?
solve(t(A)%*%A)
```

```
      [,1] [,2]
[1,]  0.75 -0.5
[2,] -0.50  0.5
```

```r
solve(t(A)%*%A)%*%t(A)%*%C
```

```
     [,1]
[1,] 2.25
[2,] 2.00
```

Let's go back to our WHI data

```r
betas<- solve(t(X)%*%X)%*%t(X)%*%whi$breastcancer
dim(betas)
```

```
[1] 4 1
```

```r
betas
```

```
                [,1]
constant 0.014945101
treat    0.005112105
age60s   0.005336953
age70s   0.008167928
```

We got the same answer as lm!

optional: make a new version of X that includes all three age variables. What happens?

```r
df.mat2<- data.frame(constant=rep(1,length(whi$breastcancer)), treat=whi$treat, age50s=whi

X2<- data.matrix(df.mat2)

dim(X2)
```

```
[1] 16608     5
```

```r
#solve(t(X2)%*%X2)
```

How do we use matrices to estimate standard errors for our betas? What is the sandwich estimator?

$$\hat{V}\left[\hat{\beta}\right] = \left[X^T X\right]^{-1} X^T \text{diag}\left[(Y - X\hat{\beta})^2\right] X \left[X^T X\right]^{-1}.$$

We know we can deal with X now, but what is the thing in the middle of the sandwich? A diagonal matrix of the residuals squared.

```r
resid<- residuals(whimod1)
```

so we have our residuals and we can square them, but how do we make a diagonal matrix?

```r
test<- c(4,5,6,7,8)

diag(test)
```

```
     [,1] [,2] [,3] [,4] [,5]
[1,]    4    0    0    0    0
[2,]    0    5    0    0    0
[3,]    0    0    6    0    0
[4,]    0    0    0    7    0
[5,]    0    0    0    0    8
```

So we can make a diagonal matrix of our squared residuals the same way. What should the dimensions of the matrix be?

```r
D<- diag(resid^2)

dim(D)
```

```
[1] 16608 16608
```

Let's use our sandwich estimator

```
betavar<- solve(t(X)%*%X)%*%t(X)%*%D%*%X%*%solve(t(X)%*%X)
```

Are we done? The diagonal is the variance of our betas and we need to take the squareroot to get the standard error.

```
sqrt(diag(betavar))
```

```
   constant       treat      age60s      age70s 
0.002061465 0.002257021 0.002471209 0.003181865 
```

That was complicated. Big matrices. Is there an easier way to do it? The sandwich package

```
library(sandwich)
sandwich(whimod1)
```

```
              (Intercept)         treat        age60s        age70s
(Intercept)  4.249638e-06 -2.401299e-06 -3.154447e-06 -3.273013e-06
treat       -2.401299e-06  5.094142e-06  5.382982e-08  2.871100e-07
age60s      -3.154447e-06  5.382982e-08  6.106872e-06  3.127476e-06
age70s      -3.273013e-06  2.871100e-07  3.127476e-06  1.012427e-05
```

```
sqrt(diag(sandwich(whimod1)))
```

```
(Intercept)       treat      age60s      age70s 
0.002061465 0.002257021 0.002471209 0.003181865 
```

```
sqrt(diag(vcovHC(whimod1, type="HC0")))
```

```
(Intercept)       treat      age60s      age70s 
0.002061465 0.002257021 0.002471209 0.003181865 
```